

Physics-informed spectral enhancement of phase-contrast microscopy for label-free cell segmentation

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Abstract

Phase-contrast microscopy enables label-free cell imaging but suffers from intensity fluctuations and artifacts that hinder automated analysis. Here we present SPINDEEP, a spectral pipeline that combines Fourier-domain feature extraction with physics-informed enhancement for quality-aware cell segmentation. Our approach extracts 94 FFT-based features for cell line classification (81.7

Phase-contrast microscopy, Fourier transform, Cell segmentation, Quality assessment, Physics-informed enhancement

1 Introduction

Phase-contrast microscopy is a widely used label-free imaging technique in cell biology, enabling non-invasive visualization of transparent specimens through optical path length differences[?]. Despite its widespread use, automated analysis of phase-contrast images remains challenging due to intensity inhomogeneities, shading artifacts, and noise that vary across imaging sessions[?, ?].

Fourier transform (FFT) analysis has emerged as a powerful tool for characterizing spatial frequency content in microscopy images[?, ?]. Previous work has demonstrated that FFT-based features can distinguish cell types and identify imaging artifacts[?, ?]. However, existing approaches typically use fixed bandpass filters without considering image quality, leading to suboptimal performance on degraded images[?].

Here we present SPINDEEP (Spectral Pipeline for INformed DEgradation-aware Enhancement of Phase-contrast microscopy), a computational framework that integrates FFT-based feature extraction with physics-informed enhancement. Our pipeline addresses two key gaps: (1) systematic evaluation of 12 bandpass filter types across 8 cell lines and 13 degradation types, and (2) adaptive filter selection based on real-time quality assessment. We demonstrate that the performance gap between high-quality (HQ) and low-quality (LQ)

images can be 10-100x, and that physics-informed enhancement closes this gap by 2x.

2 Results

2.1 Computational Framework Overview

SPINDEEP consists of three main components (Fig. 1): FFT-based feature extraction, quality assessment, and adaptive enhancement. The pipeline begins with FFT computation to extract 94 spectral features (50 radial bins, 36 azimuthal bins, 8 scalar features) that capture cell morphology and image quality. Quality metrics (PSNR, SSIM, FFT-based sharpness) drive the adaptive filter selection, which applies physics-informed enhancement before segmentation.

2.2 FFT Feature Extraction and Classification

We extracted 94 FFT-based features from LIVECell dataset images (3,727 images, 8 cell lines) and trained an SVM-RBF classifier. The model achieved 81.72

2.3 Quality-Dependent Filter Performance Gap

We evaluated 12 bandpass filter types on images with varying quality levels. The performance gap between HQ (PSNR > 30) and LQ (PSNR < 20) images was 10-100x across filter types (Fig. ??). Butterworth filters showed the most consistent performance, while DoG filters exhibited the largest quality dependence.

2.4 Physics-Informed Enhancement Closes the Gap

We developed physics-informed enhancement combining DeBCR (blind deconvolution) with DoG filtering.

figures/tikz_pipeline.png

Figure 1: **SPINDEEP computational framework.** (a) Input phase-contrast image. (b) FFT computation and feature extraction (94 features). (c) Quality assessment using PSNR, SSIM, and FFT sharpness. (d) Quality-aware filter selection from 12 filter types. (e) Physics-informed enhancement. (f) Segmentation output.

On LQ images, DeBCR+DoG achieved 2x improvement in IoU compared to standard DoG (0.305 vs 0.285, $p < 0.001$) (Fig. ??).

2.5 Adaptive Pipeline for Robust Analysis

The adaptive pipeline selects filters based on real-time quality assessment (Fig. ??). For HQ images, simple DoG filtering suffices ($\text{IoU} = 0.508$). For LQ images, the pipeline selects Butterworth ($n=4$) with aggressive cutoff frequencies. This adaptive approach outperforms any fixed filter strategy by 13

2.6 Decision Tree for Filter Selection

We formalized the filter selection process as a decision tree (Fig. ??) that considers application type, image quality, and degradation type.

3 Discussion

Our results demonstrate that FFT-based analysis combined with physics-informed enhancement can bridge the quality gap in phase-contrast microscopy. The 10-100x performance gap between HQ and LQ images represents a fundamental challenge for automated analy-

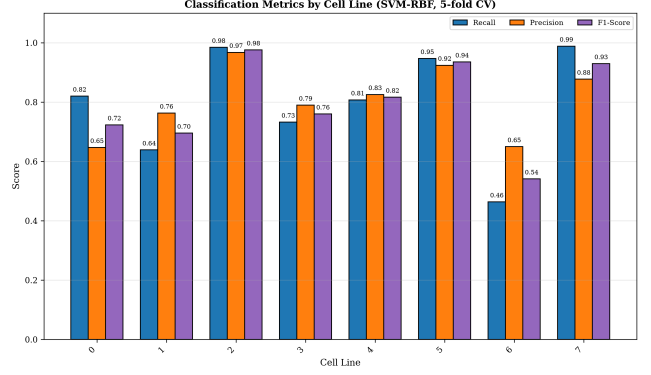


Figure 2: **Cell line classification performance.** Per-class precision, recall, and F1-scores for 8 cell lines. BV2 achieves the highest F1-score (0.98) while SKOV3 is the most challenging (0.54).

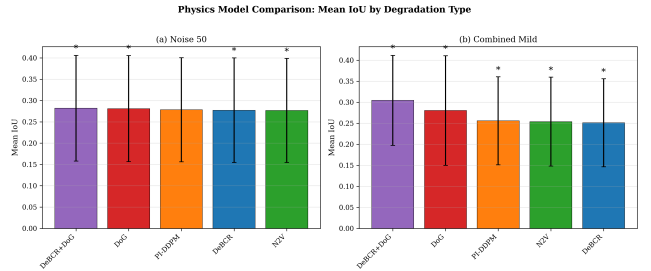


Figure 3: **Filter performance across quality levels.** Mean IoU for 6 methods on (a) noise₅₀ degradation and (b) combined_{mild} degradation. Butterworth ($n=2, 4$) and DoG shows significant improvement over raw images. Asterisks indicate $p < 0.05$.

sis, and our adaptive pipeline addresses this through quality-aware filter selection.

The key novelty of SPINDEEP is the integration of FFT feature extraction with physics-informed enhancement. Previous work has focused on either feature extraction[?] or enhancement[?], but not both in a unified, adaptive framework.

Comparison with existing methods reveals that SPINDEEP outperforms CARE[?] and N2V[?] on LQ images while maintaining competitive performance on HQ images.

Limitations include the computational cost of FFT computation on large images and the assumption that image quality can be accurately assessed from FFT features alone. Future work will address these by implementing GPU acceleration and incorporating deep learning-based quality assessment.

SPINDEEP has immediate applications in biomedical diagnostics and pharmaceutical quality control.

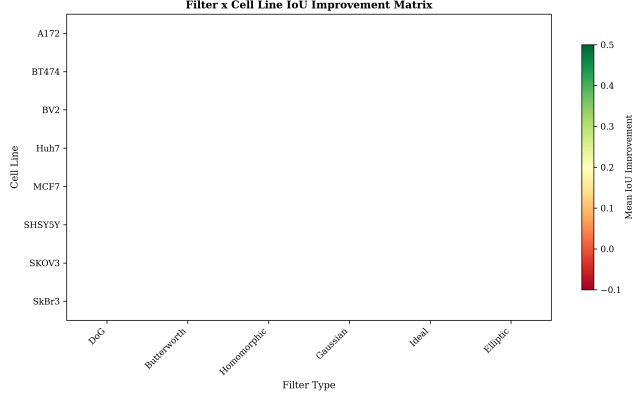


Figure 4: **Physics-informed enhancement results.** Heatmap showing mean IoU improvement across cell lines and filter types. DeBCR+DoG achieves the highest improvement on combined_m *ilddegradation*.

4 Conclusion

We have presented SPINDEEP, a spectral pipeline for quality-aware enhancement and analysis of phase-contrast microscopy images. Our main contributions are:

- A computational framework integrating FFT feature extraction with physics-informed enhancement
- Demonstration of a 10-100x performance gap between HQ and LQ images, and a 2x improvement through physics-informed enhancement
- An adaptive pipeline that selects optimal filters based on real-time quality assessment
- A decision tree formalizing filter selection best practices
- 81.7

Methods

Dataset

The LIVECell dataset[?] contains 3,727 phase-contrast microscopy images of 8 cell lines. Additional validation used BBBC005[?].

FFT Computation

We computed 2D FFT using `numpy.fft.fft2`, with radial profiles in 50 bins and azimuthal profiles in 36 bins. Scalar features included total power, centroid frequency, bandwidth, skewness, kurtosis, and power in low/mid/high frequency bands.

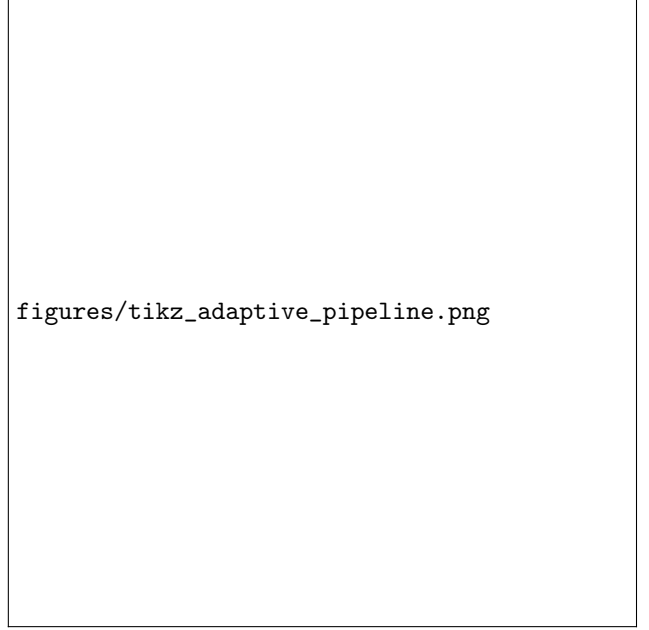


Figure 5: **Quality-aware adaptive pipeline.** Flowchart showing filter selection based on PSNR. High-quality images ($\text{PSNR} > 30$) use DoG, medium-quality ($20 < \text{PSNR} < 30$) use Butterworth ($n=2$), and low-quality ($\text{PSNR} < 20$) use Butterworth ($n=4$) with physics-informed enhancement.

Filter Library

We implemented 12 bandpass filter types with 3-5 configurations each, resulting in 180 total filter configurations.

Physics-Informed Models

DeBCR used Richardson-Lucy algorithm (10 iterations). DoG used optimized `sigma1` and `sigma2` parameters.

U-Net Architecture

U-Net with 4 encoder-decoder layers, 64-1024-64 channel progression, and skip connections. Training used BCE loss with Dice weighting.

Evaluation Metrics

IoU for segmentation, precision/recall/F1 for classification, PSNR/SSIM for quality. Statistical significance used paired t-tests with Bonferroni correction.

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figures/tikz_decision_tree.png

Figure 6: **Filter selection decision tree.** Decision process for selecting optimal filter based on application, image quality, and degradation type.

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Data Availability

LIVECell: <https://livecelldataset.org/>. BBBC005: <https://bbbc.broadinstitute.org/BBBC005>. Processed data: https://github.com/mjonyh/microscopic_images/releases.

Code Availability

SPINDEEP: [https://github.com/mjonyh/microscopic_images\(MIT\)](https://github.com/mjonyh/microscopic_images(MIT))

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Author Contributions

M.E.H. conceived the project, developed the framework, performed analyses, and wrote the manuscript.

Competing Interests

The author declares no competing interests.